



# CSE 417: Software Engineering & Design Pattern

## Lecture 1: Introduction to Software Engineering

---

Md. Mushtaq Shahriyar Rafee

January 14, 2026

Assistant Professor and Head  
Department of Data Science  
Metropolitan University

# Table of contents

1. What is Software Engineering?
2. Software Application Domains
3. System Software vs Application Software
4. Software Engineering vs Computer Engineering
5. Essential Attributes of a Good Software

# What is Software Engineering?

---

# What is Software Engineering? i

## What is a Software?

A **collection of programs** that accomplish any particular task!!

## What is Engineering?

Application of **scientific and practical knowledge** to invent, design, build, maintain, and improve frameworks, processes, etc.

# What is Software Engineering? i

What then is the definition of Software Engineering?!!



# What is Software Engineering? i

## Definition

Software engineering is an engineering discipline that is concerned with from the **early stages** of **system specification** through to **maintaining** the system after it has gone into use.



# Software Application Domains

---

## SEVEN Broad Categories!!

### System Software

- Software designed to **provide a platform for other software or service to other programs**.
- Example: Various operating systems (Android, Windows, Linux, macOS etc).

### Application Software

- **Stand-alone** programs that **solve a specific business need**.
- Example: MS Office, PowerPoint, Chrome, Adobe Photoshop, Notepad, Skype).



# Software Application Domains i

## Engineering/ Scientific Software

- Software satisfies the needs of a scientific or engineering user to perform enterprise-specific tasks.
- Example: MATLAB, AUTOCAD, PSPICE, ORCAD, etc.

## Embedded Software

- Resides within a product or system and is used to implement and control features and functions for the end user and for the system itself.
- Key pad control for a microwave oven or washing machines.

# Software Application Domains i

## Web applications

- Application software that is **accessed using a web browser**
- Provide stand-alone features, computing functions, and content to the end user, also integrated with corporate databases and business applications.
- Example: online forms, shopping carts, video and photo editing, file conversion.

## Product-line/Business Software

- Software is **used to support business applications** and is the most widely used category of software.
- Software for inventory management, accounts, banking, hospitals, schools, stock markets, etc.

## Artificial intelligence software

- Computer program which **mimics human behavior** by learning various data patterns and insights.
- Example: robotics, expert systems, pattern recognition (image and voice), artificial neural networks, theorem proving, and game playing.

# System Software vs Application Software

---

# System Software vs Application Software i

| System Software                                                                       | Application Software                                      |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Maintains <b>system resources</b> & gives <b>path for application</b> software to run | Built for specific tasks.                                 |
| <b>Low-level</b> languages are used                                                   | <b>High-level</b> languages are used                      |
| System <b>stops</b> without system software                                           | Without application software system always <b>runs</b> .  |
| Runs <b>independently</b> .                                                           | <b>Dependent</b> on system software                       |
| Operates the system in the <b>background</b> until the shutdown of the computer       | Runs in the <b>front</b> according to the user's request. |
| Example: OS                                                                           | Example: Photoshop, VLC player, etc.                      |

# Software Engineering vs Computer Engineering

---

# Software Engineering vs Computer Engineering i

## Software Engineering?

- Study of **software** which tell us about how Software is formed and about the processes involved in the formation of Software.
- Applies the principles of Engineering in order to create a software.

## Computer Engineering?

- Study of both **software and hardware** and informs about the theoretical and practical implementation of mathematical formulations and technologies.
- Provide knowledge about various field: networking, processors and data base etc.
- Base of Software Engineering.

# Essential Attributes of a Good Software

---



# Essential Attributes of a Good Software i

## Maintainability

- Software should be written in such a way so that it can **evolve** to meet the **changing needs** of customers.
- This is a critical attribute because software change is an **inevitable requirement** of a **changing business environment**.

## Dependability and security

- Software dependability includes a range of characteristics including **reliability, security, and safety**. Dependable software **should not** cause **physical or economic damage** in the event of system failure. **Malicious users** should not be able to **access or damage** the system.

# Essential Attributes of a Good Software i

## Efficiency

- Software **should not** make wasteful use of system resources such as **memory and processor** cycles. Efficiency therefore includes responsiveness, processing time, memory utilization, etc.

## Acceptability

- Software must be **acceptable** to the **type of users** for which it is designed. This means that it must be understandable, usable, and compatible with other systems that they use.

Any Questions??